Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **C++/Python systems**, **concurrent data pipelines**, and performance-focused **Linux applications** with automated testing, simulation, debugging, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **concurrent backend orchestration** with explicit job states, failure isolation, recoverable processing, and observable workflows under production load reliably daily.
- Achieved 99%+ production reliability across 5K+ monthly jobs through **data validation**, reconciliation checks, error handling, cloud storage integration, and CI/CD releases connecting FastAPI services and relational databases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Processed 1K+ files per batch using **Python multithreading**, validating PDF, DOCX, and ZIP inputs while isolating failures and preserving responsive workflows during concurrent model execution at scale.
- Reduced feedback turnaround 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows reliably.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building an **algorithmic Python pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Raised closed-loop stability 15% through **quantitative performance analysis** of HD maps, ego paths, nearby agents, attention, and distance-aware interactions across diverse autonomous-driving simulations and edge cases reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows at scale.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Reduced centralized-control and quantum-risk exposure 25% in benchmark simulations by designing **consensus algorithms** and evaluating five lattice-based cryptographic primitives across distributed protocol scenarios for peer-reviewed research publication.
- Published peer-reviewed **systems research** and presented at IISc Bengaluru, documenting architecture, simulation methodology, threat models, experimental trade-offs, and reproducible technical findings for distributed consensus design and evaluation.

Skills

Languages: C++, Python, Java, Bash, Shell, SQL, JavaScript
Systems & CS: Linux, Object-Oriented Design, Data Structures, Algorithms, Concurrency, Multithreading, Process Scheduling, Distributed Systems, TCP/IP, JSON
Testing & Performance: Unit Testing, Integration Testing, PyTest, Mockito, Debugging, Performance Optimization, Simulation, Data Validation, Error Handling
Tooling & Delivery: Git, GitHub, GitHub Actions, Jenkins, CI/CD, Docker, Kubernetes, AWS, Azure, GCP
Data & Engineering: REST APIs, PostgreSQL, ETL Pipelines, System Design

Projects

Multi-Sensor Data Acquisition Pipeline - Design Concept

- Designed a modern **C++20 multi-sensor pipeline** concept using producer-consumer queues, bounded buffers, timestamp alignment, binary storage, backpressure, and deterministic shutdown for reliable Linux acquisition and processing workflows.

Hardware-in-the-Loop Diagnostics Harness - Design Concept

- Designed a **C++/Python hardware-in-the-loop harness** concept supporting simulated and physical sensors, USB/serial/Ethernet adapters, fault injection, log capture, repeatable test scenarios, GDB debugging, and repeatable performance profiling workflows.

CPU Scheduling Simulator | Live

- Implemented **C++ scheduling algorithms** for FCFS, SJF, SRT, Round Robin, priority, and MLQ, with edge-case handling, process-state visualization, and dynamic turnaround and waiting-time calculations across simulations reliably.